

InfDB: An Open Source Energy and Infrastructure Data Ecosystem for Modeling and Planning

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Summary

InfDB – Energy and Infrastructure Database is an open source, containerized data infrastructure for managing and providing access to heterogeneous energy and infrastructure datasets used in urban and regional energy system studies.

It bundles a PostgreSQL ([Stonebraker & Rowe, 1986](#)) database extended for geospatial, time-series, and graph data as well as standardized REST ([Fielding, 2000](#)) and OpenAPI standards by the Open Geospatial Consortium (OGC) ([Open Geospatial Consortium, 2010](#)). Finally, it offers a configurable import service transforming raw public data into structured, version-controlled schemas.

By decoupling data ingestion, storage, and access from downstream analysis, modeling and planning tools, InfDB reduces data preprocessing effort and enables reproducible, transferable energy system modeling workflows across regions and projects. In this way, it can be used for many different applications in energy system modeling, such as district heating planning, electrical distribution network analysis, or urban energy demand estimation.

Statement of need

The transition to a climate-neutral energy system is a central pillar of energy policy, exemplified by Germany's aim for climate neutrality by 2045. New legislative frameworks, such as the requirement for municipal heat planning ([Wärmeplanungsgesetz - WPG, 2023](#)) and the requirement for grid expansion plans based on regional transition pathway scenarios defined in the German Energy Industry Act ("Gesetz über die Elektrizitäts- und Gasversorgung") ([Gesetz Über Die Elektrizitäts- Und Gasversorgung, 2026](#)), demand that municipalities and Distribution System Operators (DSOs) process vast amounts of energy and infrastructure data.

However, the current landscape of energy data is fragmented. While the Open Data Strategy of the German Federal Government ([Bundesministerium des Innern für Bau und Heimat, 2021](#)) has increased data availability, this data is published by disparate authorities on different platforms in varying formats, spatial resolutions, and licensing structures. Consequently, energy modeling workflows often suffer from:

1. **High Preprocessing Effort:** Researchers and professionals spend a disproportionate amount of time acquiring raw data before analysis.
2. **Limited Workflow Transferability:** Data processing workflows often require substantial adaptation across regions due to differing data formats, interfaces, and conventions.

3. **Lack of Reproducibility:** Planning results are often one-off studies that are difficult to update or audit.
4. **Siloed Infrastructure:** DSOs and municipalities lack standardized tools to efficiently manage and share energy and infrastructure data, such as distribution network or building data.

InfDB addresses these challenges by providing a reproducible, version-controlled, and automated ETL (Extract, Transform, Load) pipeline. It acts as a middleware between raw public data and high-level energy modeling and planning tools, ensuring that input data is transparent, traceable, and easily updatable. While InfDB is developed in a German context addressing local regulatory requirements, the underlying architecture and methodology can also be applied to data across different regions and regulatory frameworks, making it applicable to municipal and regional energy planning efforts worldwide.

State of the field

Energy and infrastructure data management is an active field with several existing solutions:

- **Commercial Platforms:** Tools like **nPro** (Wirtz, 2023), **Solarea** (Solera GmbH, 2026), and **flexRM** (netCADservice GmbH, 2026) offer robust analytics and user interfaces but are proprietary, limiting transparency and community extension.
- **Open Source Modeling Frameworks:** Tools like **City Energy Analyst (CEA)** (Fonseca et al., 2025), **EURECA** (Prataviera et al., 2021) and **OpenPlan** (OpenPlan, 2020) focus on modeling and optimization but typically assume preprocessed, structured input data provided externally.
- **Data Initiatives:** The **NEED** project (Duchon et al., 2024) provides a decentralized data hub for synthetic energy data, and **DB4KWP** (Reiner-Lemoine-Institut, 2026) focuses on ontologies Open Energy Ontology (OEO) and Knowledge Graph (OEKG) naming conventions.

InfDB fills the gap between available open data sources and existing energy modeling and planning solutions. While simulation tools like CEA focus on modeling, InfDB provides the foundational data infrastructure that these tools require. Moreover, unlike static data repositories, InfDB offers a dynamic, service-oriented platform that enables users to deploy local instances, continuously integrate fresh datasets, and seamlessly connect with both commercial and open source downstream tools. Looking forward, InfDB can complement ontology initiatives like DB4KWP by providing a technical implementation layer that transforms conceptual data standards into practical, operational systems.

Research impact statement

The research relevance of InfDB lies in its role as a reusable data infrastructure that supports transparent and reproducible energy system analysis workflows. By separating data management from analysis logic, InfDB contributes to several recurring methodological requirements in energy research:

- **Reproducibility:** Containerized deployment and configuration-based data ingestion allow complete data pipelines to be rerun and inspected. This enables studies to be reproduced or updated when new data becomes available, without reimplementing preprocessing steps.
- **Transferability:** By enforcing structured schemas and standardized access interfaces, InfDB allows data processing and analysis workflows to be reused across regions and projects with minimal adaptation, reducing region-specific reimplementation effort.
- **Separation of Concerns:** InfDB decouples data acquisition, storage, and access from modeling and analysis code. This allows researchers to develop and test analytical

methods independently of changes in input data formats or sources.

- **Workflow Stability:** Stable database schemas and interfaces provide a consistent foundation for iterative research, supporting comparative studies and sensitivity analyses without requiring repeated adjustments to data handling logic.
- **Methodological Neutrality:** InfDB does not prescribe modeling approaches, optimization methods, or policy assumptions. Its role is limited to providing structured and accessible data, allowing a wide range of analytical methods to be applied without constraint.

Software Design

InfDB is designed as a modular, containerized data infrastructure that decouples data ingestion, storage, and access from downstream analysis and modeling. It follows a service-oriented architecture orchestrated via Docker Compose, allowing individual components to be deployed, configured, and combined depending on the requirements of a specific workflow. The system is conceptually divided into **Services**, which provide the foundational infrastructure, and **Tools**, which consume and process the data. As opposed to a single binary or service, this separation allows for data portability between tools and configurability both in deciding which components are enabled and in easily expanding to new components or replacing components to meet different preferences or local needs.

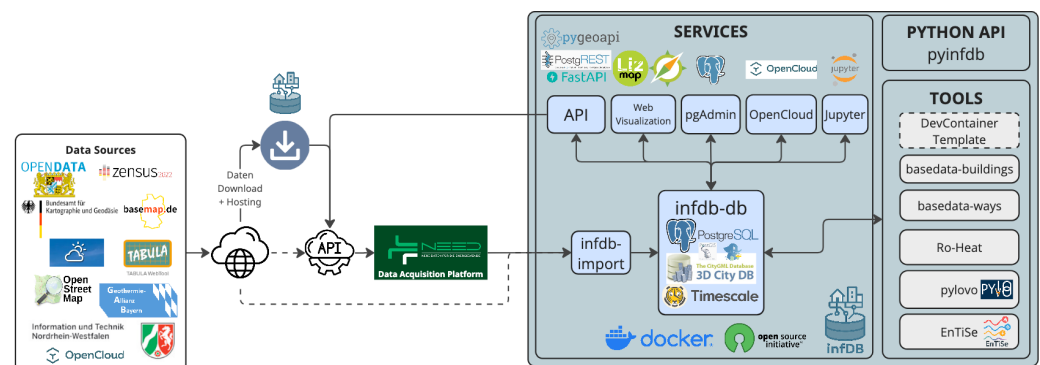


Figure 1: InfDB - Data Sources, Services and Tools

Services

The *Services* layer (depicted in the grey box in Figure 1) handles database operations, administration, data ingestion, and connectivity. These containerized services include:

infdb-import

This service automates the ingestion of heterogeneous external data sources. It transforms raw external formats into structured schemas within the database. Users control this process via a simple YAML configuration file, eliminating the need for custom ETL scripting.

infdb-db

The central storage engine hosting a PostgreSQL (PostgreSQL Global Development Group, 2026) database. It is pre-configured with essential extensions for infrastructure and energy data:

- **PostGIS** (PostGIS Project Steering Committee, 2026) for geospatial data.
- **TimescaleDB** (Timescale, Inc., 2026) for time-series data.
- **3D City DB** (Yao et al., 2018) for (3D) semantic city models.
- **pgRouting** (pgRouting Community, 2026) for graph-based network analysis.

APIs and Data Access Services

InfDB exposes data exclusively through standardized interfaces rather than custom file formats. This includes SQL access to the database as well as RESTful and OGC-compliant APIs implemented using **FastAPI** (Ramírez, 2018), **PostgREST** (PostgREST Contributors, 2026), and **pygeoapi** (pygeoapi Contributors, 2026). These interfaces allow external tools to access data in a consistent manner while keeping the internal database structure encapsulated.

Administrative and Interactive Services

Optional services such as **pgAdmin** (pgAdmin Development Team, 2026), **Jupyter Notebook** (Kluyver et al., 2016), and a web-based **QGIS** (QGIS Development Team, 2026) client support administration, inspection, prototyping, and visualization of data stored in the database. These components are intended to lower the entry barrier for users from different backgrounds (e.g., GIS specialists or researchers). However, they are not required for automated or headless workflows. An external storage integration, such as the **OpenCloud** (OpenCloud GmbH, 2026) component, allows integration with cloud-based storage solutions for handling offline or closed data.

Tools

The *Tools* layer (depicted in the right box in Figure 1) consists of (external) software that interacts with the InfDB Services to process data or generate insights. Each tool can interact independently with the Services, reading and writing data. This modular approach allows users to chain different tools into custom workflows. Depending on the tool type and requirements, e.g., new scripts, open source tools, proprietary tools, multiple integration options with InfDB are available building upon the following foundations:

- **Standardized Integration:** Tools interact with the core database exclusively through open interfaces (SQL or REST APIs), ensuring that the underlying data schema remains consistent regardless of the tool used.
- **pyinfdb:** To facilitate the development of custom tools, the platform provides the `pyinfdb` Python package. This library abstracts database connections, logging, and configuration management, allowing researchers to rapidly develop Python-based analysis scripts that integrate seamlessly with the InfDB ecosystem.
- **Extensible Ecosystem:** Importantly, InfDB does not prescribe specific modeling approaches, optimization methods, or planning workflows. Its role is limited to providing a stable and reproducible data infrastructure that can be reused across different analytical contexts. Therefore, users can integrate existing third-party simulation software or develop proprietary tools that plug into the InfDB backend without modifying the core services.

Applications

The InfDB ecosystem allows researchers and planners across multiple use cases to access the full spectrum of container-ingested data. This includes open data (e.g., building LOD2), enriched base data (e.g., estimated households per building), and intermediate simulation results (e.g., heat demand). This also facilitates the seamless integration of downstream tools, such as energy system optimization models, with consistent data for both electrical and district heating grids. This makes InfDB suitable both for one-time feasibility studies and for continuous planning processes where data and assumptions are updated regularly.

AI usage disclosure

In the development of this work, GitHub Copilot (GitHub, Inc., 2026) and Gemini (Google LLC, 2026) were used with human oversight. GitHub Copilot Chat was used in software code and project documentation for code completion, refactoring suggestions, and boilerplate generation. Gemini 1.5 Pro was used for paper-text drafting support, language editing, and restructuring suggestions. All AI-generated suggestions were reviewed, edited, and validated by the authors before inclusion.

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